Elementary Fluid Mechanics and Lab.
Final Exam

건설환경공학부 2022-14407 권영빈

1 Calculating ω Using the Newton-Raphson Method

```
import numpy as np

def find_omega(w, Dw, initial, k, h, tol=1e-7, max_iter=1000):
  x = initial
  iterations = 0

while abs(w(x)) > tol and iterations < max_iter:
  x = x - w(x) / Dw(x)
  iterations += 1

if iterations == max_iter:
  print("Max iterations reached, may not have converged.")

return x, iterations

if __name__ == "__main__":

  def W(x):
    return g * k * np.tanh(k * h) - x ** 2

  def DW(x):
    return (-2) * x

root, iterations = find_omega(W, DW, initial, k, h)
print(f"Omega = {root}, No. of Iterations = {iterations}")
```

We simply define function `find_omega` that takes functions `w` and `Dw` as input, along with initial values of k and h. `tol` stands for tolerance, while `max_iter` indicates the maximum number of allowed iterations, which we initially set as `1000`. We define `W(x)` and `DW(x)` and pass it to the function. For example, we set initial value as follows:

```
initial = 2.0
k = 2.5
h = 100
g = 9.81
```

and obtain `Omega =4.952272205778136, No. of Iterations =5`.

2 Plotting as a Function of *kh*

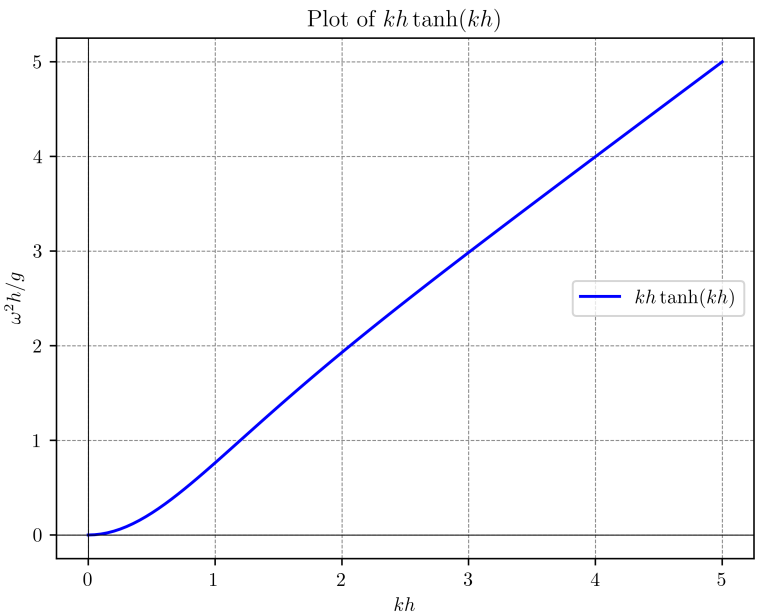

Fig. 1

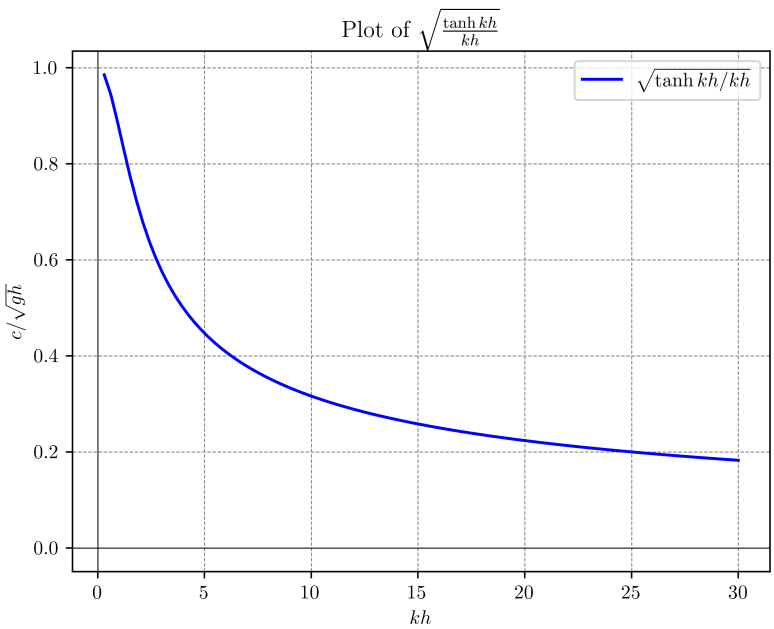

Fig. 2

3 Pressure from the Unsteady Bernoulli Equation

3.1 Unsteady Bernoulli Equation

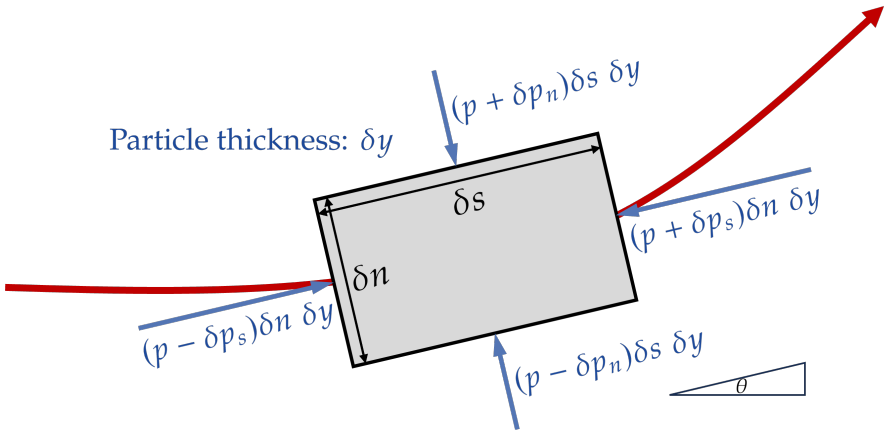

Fig. 3: Small Fluid Particle Along a Streamline

We first obtain the unsteady Bernoulli equation. We consider a small fluid particle along a streamline of size δs by δn in the plain, and denote forces exerted to the particle as above in Fig. 3. δp_s and δp_n are respectively amounts of the equal pressure differences of end faces parallel to the streamline and perpendicular to the streamline. We hereby write Newton's second law along the streamline direction as follows:

$$\Sigma\, \delta F_s = \delta m\, a_s, \tag{1}$$

where a_s denotes acceleration along the streamline. We know that velocity $V(s,t)$ is a function of location in the streamline s and time t. Accordingly, we write

$$\Sigma\, \delta F_s = \delta m \left(\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial s} \right) = \rho\, \delta V_{\text{control}} \left(\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial s} \right).$$

Meanwhile, $\Sigma\, \delta F_s = \delta W_s + \delta F_{ps}$ where δW_s is gravitational force along the streamline and δF_{ps} is pressure force along the streamline. We obtain δW_s and δF_{ps} from force relations as follows:

$$\begin{aligned} \delta W_s &= -\delta W \sin\theta = -\gamma\, \delta V_{\text{control}} \sin\theta \\ \delta F_{ps} &= (p - \delta p_s)\, \delta n\, \delta y - (p + \delta p_s)\, \delta n\, \delta y = -2\, \delta p_s\, \delta p_n\, \delta y. \end{aligned}$$

Here, since we are considering a small particle, we can neglect higher order terms from the pressure field, therefore $2\, \delta p_s \approx (\partial p / \partial s)\, \delta s$. Inserting this result to δF_{ps} and Eq. (1) gives

$$\Sigma\, \delta F_s = \delta W_s + \delta F_{ps} = \left(-\gamma \sin\theta - \frac{\partial p}{\partial s} \right) \delta V_{\text{control}} = \rho \left(\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial s} \right). \tag{2}$$

Now integrating Eq. (2) gives us the unsteady Bernoulli equation:

$$p + \frac{1}{2}\rho V^2 + \gamma z + \rho \frac{\partial \phi}{\partial t} = 0. \tag{3}$$

3.2 Pressure from the Linearized Unsteady Bernoulli Equation

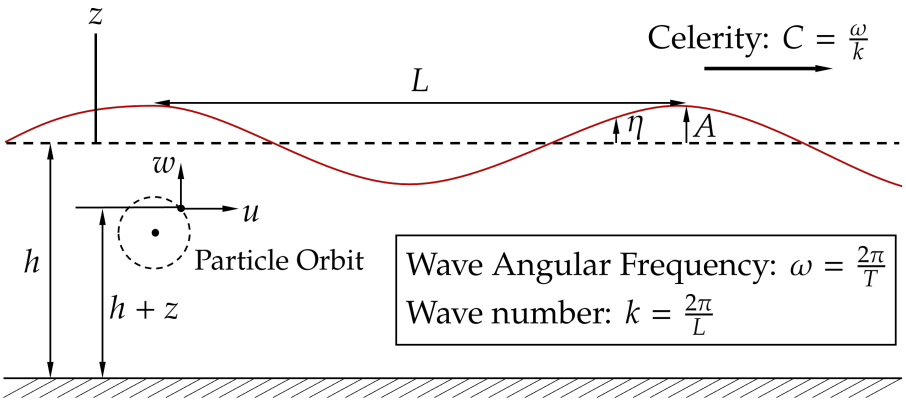

Fig. 4: Free Surface Waves

We next apply the unsteady Bernoulli equation according to the notations given in Fig. 4 above:

$$\frac{1}{2}(u^2 + w^2) + \frac{P}{\rho} + gz + \frac{\partial \Phi}{\partial t} = 0, \tag{4}$$

where u and w are each horizontal and vertical components of the water particle velocity and h is water depth.

Here, we linearize and omit higher-order terms, then write P as follows:

$$P = -\rho g z - \rho \frac{\partial \Phi}{\partial t}. \tag{5}$$

Then we calculate $\partial\Phi/\partial t$:

$$\frac{\partial \Phi}{\partial t} = -gA\left[\frac{\cosh k(z+h)}{\cosh kh} e^{i(kx-\omega t)}\right]. \tag{6}$$

Note that Φ here is a complex potential $\Phi = \phi + i\psi$, where

$$\phi = \frac{gA}{\omega}\frac{\cosh k(h+z)}{\cosh kh}\sin(kx - \omega t),$$

and ψ is its complex conjugate potential. Now we insert Eq. (6) to Eq. (5). We here insert the real part of the derivative, therefore obtain:

$$P = -\rho g z + \rho g A\left[\frac{\cosh k(z+h)}{\cosh kh}\right]\cos(kx - \omega t). \tag{7}$$

Here, we see that the first term on the right $-\rho g z$ gives us the normal hydrostatic pressure variation relative to z while the second term indicates the dynamic pressure from particle acceleration induced by the wave.

4 Wave Train Propagation Over a Uniform Slope

4.1 Plotting Wave Transformation

We first consider the group velocity of wave trains. Considering consecutive waves with phases δ_1 and δ_2 respectively, we may write the composite wave of the two waves as follows:

$$\begin{aligned}\eta_T = \eta_1 + \eta_2 &= A\cos(kx - \omega t + \delta_1) + A\cos(kx - \omega t + \delta_2)\\ &= 2A\cos\left[\frac{1}{2}(k_1 - k_2)x - \frac{1}{2}(\omega_1 - \omega_2)t\right]\sin\left[\frac{1}{2}(k_1 + k_2)x - \frac{1}{2}(\omega_1 + \omega_2)t\right].\end{aligned}$$

Now we see that at $t = 0$, nodes occur at

$$x = \frac{\pi}{k_1 - k_2}, \frac{3\pi}{k_1 - k_2}, \cdots$$

thus we define group velocity C_G as follows:

$$C_g = \frac{\mathrm{d}x_{\text{node}}}{\mathrm{d}x} = \frac{\omega_1 - \omega_2}{k_1 - k_2}.$$

As ω_1 approaches ω_2,

$$C_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = C - L\frac{\mathrm{d}C}{\mathrm{d}L} \tag{8}$$

$$= \frac{C}{2}\left[1 + \frac{2kh}{\sinh 2kh}\right]. \tag{9}$$

Now, we derive the energy of a wave. We first calculate the kinetic energy for a unit width of wave crest and for one wave length, E_k. We calculate as follows:

$$E_k = \int_0^L \int_{-d}^0 \frac{1}{2}\rho \, \mathrm{d}x \, \mathrm{d}z (u^2 + w^2) = \frac{\rho g H^2 L}{16}. \tag{10}$$

Now, if we subtract the potential energy of a mass of still water with respect to the bottom from the potential energy of the wave form, this gives the potential energy per unit wave crest width and for one wavelength, E_p.

$$E_p = \int_0^L \rho g (h + \eta) \left(\frac{h + \eta}{2}\right) \mathrm{d}x - \rho g L d \left(\frac{d}{2}\right) = \frac{\rho g H^2 L}{16}. \tag{11}$$

From Eqs. (10) and (11), we obtain total energy in a wave per unit crest width E:

$$E = E_k + E_p = \frac{\rho g H^2 L}{8}. \tag{12}$$

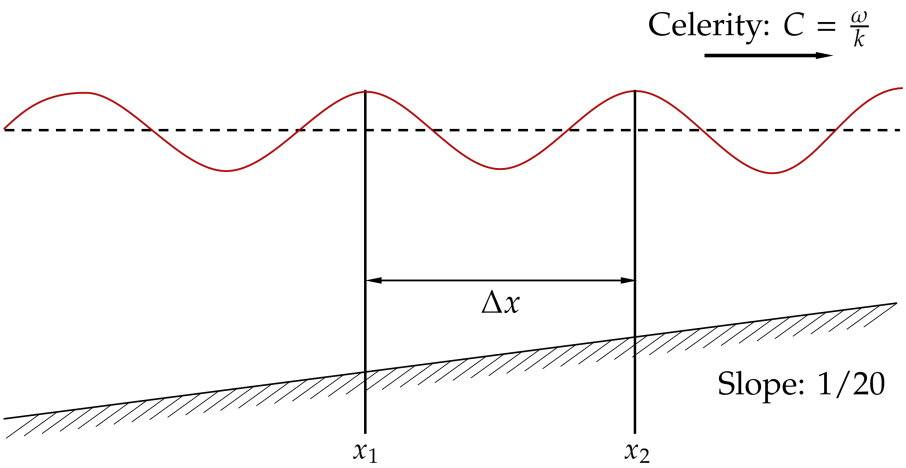

Fig. 5: Wave Propagation Towards the Shoreline

Assuming that over the distance Δx, regardless of its length, there is no generation, reflection, and dissipation of wave energy,

$$(EbC_G)_1 = (EbC_G)_2,$$

where b is the width of the channel.

From Eq. (12), we alternatively write as follows:

$$\frac{A^2 b}{4}\left[\frac{g}{k}\tanh kh\right]^{1/2}\left[1+\frac{2kh}{\sinh 2kh}\right] = const. \tag{13}$$

According to the fact that $C^2 = \frac{g}{k}\tanh kh$, we may write

$$\frac{C}{C_0} = \frac{L}{L_0} = \frac{k_0}{k} = \tanh kh, \tag{14}$$

where the subscript $_0$ indicates the value of the variable in deep water. For instance,

$$C_0 = \frac{gT}{2\pi} = 15.61\,\text{m/s} \qquad L_0 = \frac{gT^2}{2\pi} = 156.13\,\text{m}. \tag{15}$$

Meanwhile, from Eq. (13), we also obtain the relation

$$\frac{H}{H_0} = \left[\frac{b_0}{b}\right]^{1/2}\left[\frac{2\cosh^2 kh}{2kh+\sinh 2kh}\right]^{1/2}. \tag{16}$$

Now combining the two relations, and assuming constant width of channel $b = b_0$, we obtain an expression for the transformation of wave steepness $kA = \pi H/L$:

$$(kA) = (kA)_0\left[\frac{2\cosh^2 kh}{2kh+\sinh 2kh}\right]^{1/2}\coth kh. \tag{17}$$

Finally, we plot the desired physical variables in terms of h/L_0, in order to provide a practical calculation method appliable for any h. From the relation

$$kh = \frac{2\pi h}{L} = \frac{h}{L_0}2\pi\frac{L_0}{L} = \left(\frac{h}{L_0}\right)2\pi\frac{1}{\tanh kh}, \tag{18}$$

we easily substitue kh in terms of h/L_0. Plots obtained are as follows.

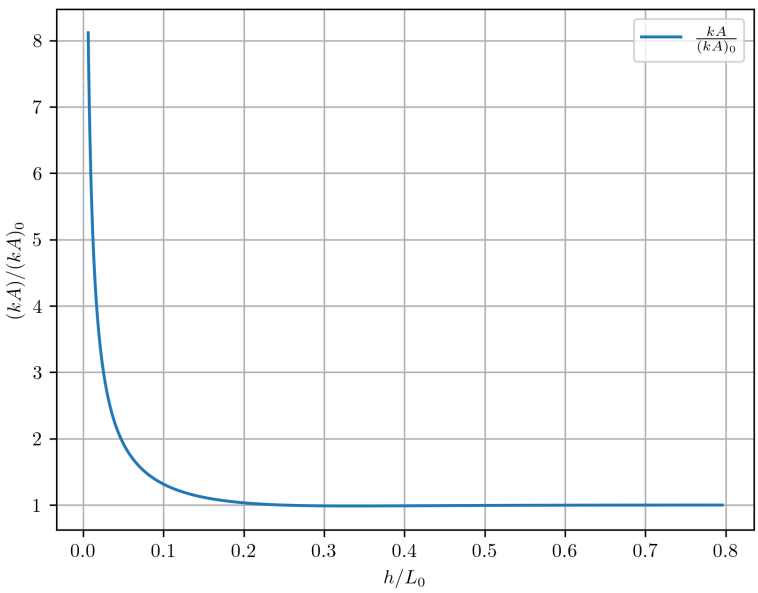

Fig. 6: Wave Steepness Transformation

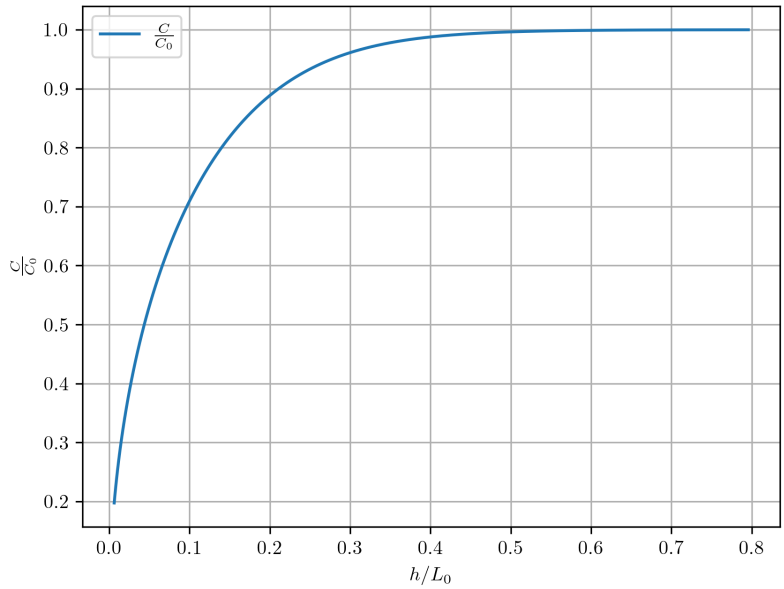

Fig. 7: Celerity Transformation

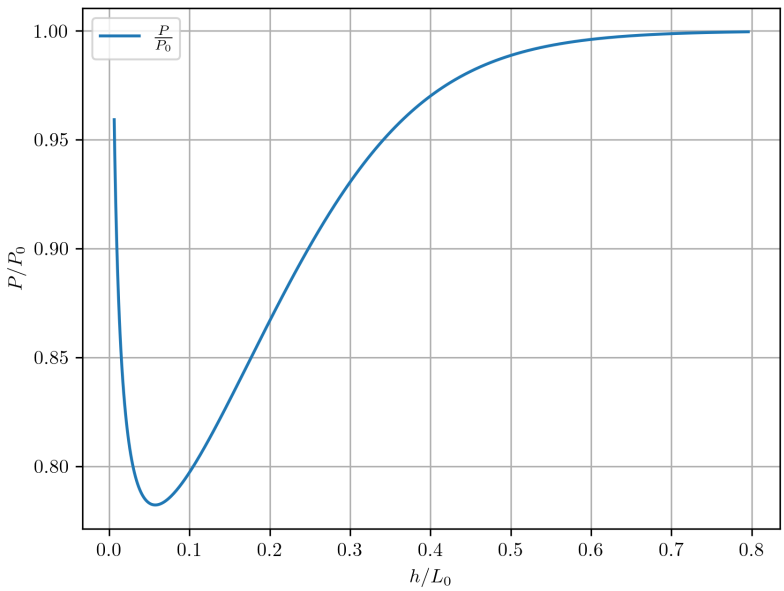

Fig. 8: Dynamic Pressure Tranformation

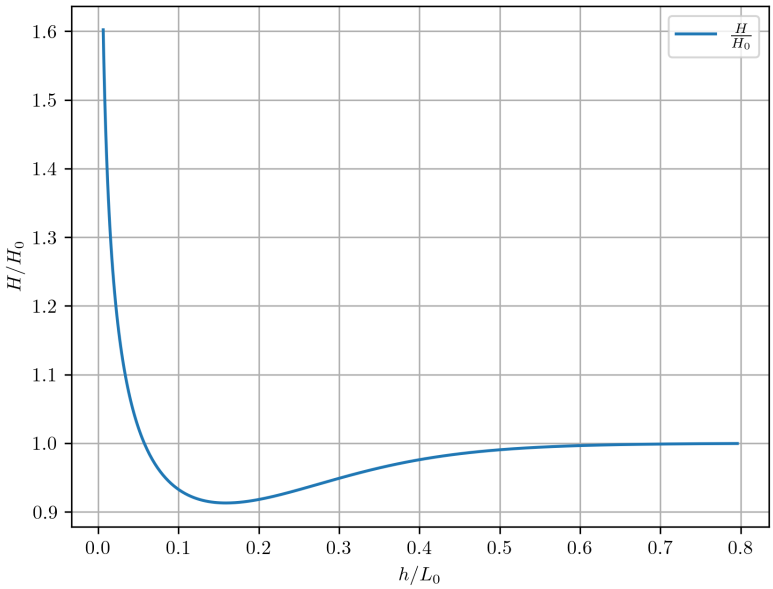

Fig. 9: Amplitude Transformation

4.2 Observations of the Plots

1. We realize that wave steepness kA gets significantly higher when approaching the shore, which in reality would break before reaching a certain point, as wave becomes unstable.

2. C/C_0 is proportional to tanh kh, as shown in the plot.

3. We examine that the ratio P/P_0 converges in deep water, while reaching a certain minimum as reaching the shore, then rises again.

4. Amplitude transformation is as Fig. 9, which also reaches minimum in $h/L_0 = 0.157$, $H/H_0 = 0.91$. As mentioned, wave breaks before amplitude rises infinitely.

For slope of this instance, the small amplitude theory is satisfactory for the prediction of phase velocity, until the point of wave breaking.

In shallow water, the wave speed depends on the water depth, and waves with different frequencies travel at different speeds, bound to the dispersion relation. As waves propagate into shallower water, the longer wavelengths travel faster than the shorter wavelengths, leading to a distortion of the wave shape. Non-linear effects, such as wave steepening and wave breaking, become more pronounced.

For accurately modeling wave behavior in shallow water, more complex models that consider non-linear effects, dispersion, and interactions with the seabed are often employed.

While the linear assumption becomes less accurate as waves enter shallow water, for a more precise analysis of wave behavior in shallow water, it is necessary to consider non-linear effects.